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|---|---|---|
| **Use case** | Registration | |
| **Release (id)** | UC1, v0.1, 19/04/2026 | |
| **Context of Use** | User authentication and account management. | |
| **Scope** | Let an anonymous user create a new account. | |
| **Level** | Primary task | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to create an account |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be logged out. | |
| **Minimal Guarantees** | No duplicated or unsecure account is created. | |
| **Success Guarantees** | The anonymous user successfully created an account and is logged in. | |
| **Trigger** | The anonymous user selects “register”. | |
| **Description** | **Step** | **Action** |
| | 1 | The anonymous user selects the functionality “register”. |
| | 2 | The system asks the user to insert username and email. |
| | 3 | The user inserts the required data. |
| | 4 | The system checks whether an account with at least one between username and email already exists. |
| | 5 | The system asks the user to insert the password twice. |
| | 6 | The user inserts the required data. |

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| | 7 | The system checks whether the password satisfies the security requirements and if the two passwords coincide. |
| | 8 | The system asks the user's consent for personal data processing. |
| | 9 | The anonymous user accepts the personal data processing. |
| | 10 | The system saves the data in the account archive. |
| | 11 | The system notifies the user of successful registration with an email. |
| | 12 | The system switches to the authenticated user functionalities. |
| **Extension** | **Step** | **Other action** |
| | 4a | Invalid username or email: UC1.1 |
| | 7a | Invalid password: UC1.2 |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Alternative flow** | Registration: Invalid username or email | |
| **Release (id)** | UC1.1, v0.2, 25/04/2026 | |
| **Context of Use** | User authentication and account management. | |
| **Scope** | The system alerts the anonymous user that they have inserted an invalid username or email. | |
| **Level** | Subfunction | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to create an account |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted an already existing username or email. | |
| **Minimal Guarantees** | No duplicated account is created. | |
| **Success Guarantees** | None | |
| **Trigger** | UC1 | |
| **Description** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid username or email. |
| | 2 | The sequence of events starts at step 2 of the main flow (UC1). |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Alternative flow** | Registration: Invalid password | |
| **Release (id)** | UC1.2, v0.2, 25/04/2026 | |
| **Context of Use** | User authentication and account management. | |
| **Scope** | The system alerts the anonymous user that they have inserted an invalid password. | |
| **Level** | Subfunction | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to create an account |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide. | |
| **Minimal Guarantees** | No unsecure account is created. | |
| **Success Guarantees** | None | |
| **Trigger** | UC1 | |
| **Description (Main Success Scenario)** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid password, specifying the problem. |
| | 2 | The sequence of events starts at step 5 of the main flow (UC1). |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | Account deletion | |
| **Release (id)** | UC2, v0.1, 19/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | Let an authenticated user delete her account. | |
| **Level** | Primary task | |
| **Primary actor** | Passenger or train company | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Passenger | Wants to delete her account |
| | Train company | Wants to delete its account |
| | System | Managse the user interface, the database and their interactions |
| **Preconditions** | The user must be logged in. | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The user successfully deleted her account and is logged out. | |
| **Trigger** | The user selects “delete account”. | |
| **Description** | **Step** | **Action** |
| | 1 | The anonymous user selects the functionality “delete account”. |
| | 2 | The system removes the data in the account archive. |
| | 3 | The system notifies the user of successful deletion with an email. |
| | 4 | The system switches to the unauthenticated user functionalities. |
| **Extension** | **Step** | **Other action** |
| | - | - |

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| **Technology & data Variations** | None |

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|---|---|---|
| **Use case** | Account data view | |
| **Release (id)** | UC3, v0.2, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | Let an authenticated user view her account's data. | |
| **Level** | Primary task | |
| **Primary actor** | Authenticated user (passenger, train company or network manager) | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Authenticated user | Wants to view her account's data |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be logged in. | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The user successfully views her account's data. | |
| **Trigger** | The anonymous user selects "view account's data". | |
| **Description** | **Step** | **Action** |
| | 1 | The anonymous user selects the functionality "view account's data". |
| | 2 | The system shows the data saved in the account archive, that is username and email. |
| **Extension** | **Step** | **Other action** |
| | - | - |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | Account data change | |
| **Release (id)** | UC4, v0.2, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | Let an authenticated user change her account's data. | |
| **Level** | Primary task | |
| **Primary actor** | Authenticated user (passenger, train company or network manager) | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Authenticated user | Wants to change her account's data |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be logged in. | |
| **Minimal Guarantees** | No duplicated or unsecure account is created with a modification. | |
| **Success Guarantees** | The authenticated user successfully changed her account's data. | |
| **Trigger** | The authenticated user selects "change data". | |
| **Description** | **Step** | **Action** |
| | 1 | The authenticated user selects the functionality "change data". |
| | 2 | The system asks the user which data to modify. |
| | 3 | If the user selects username |
| | 3.1 | The system asks the user to insert the new username. |
| | 3.2 | The user inserts the new username. |
| | 3.3 | The system checks whether an account with the same username already exists. |

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| | 4 | If the user selects email |
| | 4.1 | The system asks the user to insert the new email. |
| | 4.2 | The user inserts the new email. |
| | 4.3 | The system checks whether an account with the same email already exists. |
| | 5 | If the user selects password |
| | 5.1 | The system asks the user to insert the password twice. |
| | 5.2 | The user inserts the required data. |
| | 5.3 | The system checks whether the password satisfies the security requirements and if the two passwords coincide. |
| | 6 | The system saves the data in the account archive. |
| | 7 | The system notifies the user of successful data change with an email. |
| **Extension** | **Step** | **Other action** |
| | 3.3a | Invalid username or email: UC4.1 |
| | 4.3a | Invalid email: UC4.2 |
| | 5.3a | Invalid password: UC4.3 |
| **Technology & data Variations** | None | |

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| **Alternative flow** | Account data change: Invalid username | |
| **Release (id)** | UC4.1, v0.2, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | The system alerts the authenticated user that they have inserted an invalid username. | |
| **Level** | Subfunction | |
| **Primary actor** | Authenticated user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Authenticated user | Wants to change username |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted an already existing username. | |
| **Minimal Guarantees** | No duplicated account is created. | |
| **Success Guarantees** | None | |
| **Trigger** | UC4 | |
| **Description** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid username. |
| | 2 | The sequence of events starts at step 3.1 of the main flow (UC4). |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Alternative flow** | Account data change: Invalid email | |
| **Release (id)** | UC4.2, v0.2, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | The system alerts the authenticated user that they have inserted an invalid email. | |
| **Level** | Subfunction | |
| **Primary actor** | Authenticated user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Authenticated user | Wants to change email |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted an already existing email. | |
| **Minimal Guarantees** | No duplicated account is created. | |
| **Success Guarantees** | None | |
| **Trigger** | UC4 | |
| **Description** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid email. |
| | 2 | The sequence of events starts at step 4.1 of the main flow (UC4). |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Alternative flow** | Registration: Invalid password | |
| **Release (id)** | UC4.3, v0.2, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | The system alerts the authenticated user that they have inserted an invalid password. | |
| **Level** | Subfunction | |
| **Primary actor** | Authenticated user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Authenticated user | Wants to change password |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide. | |
| **Minimal Guarantees** | No unsecure account is created. | |
| **Success Guarantees** | None | |
| **Trigger** | UC4 | |
| **Description (Main Success Scenario)** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid password, specifying the problem. |
| | 2 | The sequence of events starts at step 5.1 of the main flow (UC4). |
| **Technology & data Variations** | None | |

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| **Use case** | Reset password | |
| **Release (id)** | UC5, v0.1, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | Let an anonymous user reset the password. | |
| **Level** | Primary task | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to reset the password |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be logged out. | |
| **Minimal Guarantees** | No unsecure account is created with a modification.<br>No user can change the password of another user. | |
| **Success Guarantees** | The anonymous user successfully changed the password. | |
| **Trigger** | The anonymous user selects “forgot password”. | |
| **Description** | **Step** | **Action** |
| | 1 | The anonymous user selects the functionality “forgot password”. |
| | 2 | The system asks the user to insert the email associated to her account. |
| | 3 | The user inserts the email. |
| | 4 | The system checks if the email is associated to an existing account. |
| | 5 | The system creates a random identifier. |
| | 6 | The system sends the identifier to the email. |
| | 7 | The system asks the user to insert the identifier it sent. |

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| | 8 | The user inserts the identifier. |
| | 9 | The system checks whether the identifier is correct. |
| | 10 | The system asks the user to insert the password twice. |
| | 11 | The user inserts the required data. |
| | 12 | The system checks whether the password satisfies the security requirements and if the two passwords coincide. |
| | 13 | The system saves the data in the account archive. |
| | 14 | The system notifies the user of successful password change with an email. |
| **Extension** | **Step** | **Other action** |
| | 4a | Invalid email: UC5.1 |
| | 9a | Invalid identifier: UC5.2 |
| | 12a | Invalid password: UC5.3 |
| **Technology & data Variations** | None | |

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| **Alternative flow** | Reset password: Invalid email | |
| **Release (id)** | UC5.1, v0.1, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | The system alerts the authenticated user that they have inserted an invalid email. | |
| **Level** | Subfunction | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to reset password |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted an email that does not correspond to any existing account. | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | None | |
| **Trigger** | UC5 | |
| **Description** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid email. |
| | 2 | The sequence of events starts at step 2 of the main flow (UC5). |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Alternative flow** | Reset password: Invalid identifier | |
| **Release (id)** | UC5.2, v0.1, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | The system alerts the authenticated user that they have inserted an invalid identifier. | |
| **Level** | Subfunction | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to reset password |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted an identifier that does not correspond to the one created by the system. | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | None | |
| **Trigger** | UC5 | |
| **Description** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid identifier. |
| | 2 | The sequence of events starts at step 9 of the main flow (UC5). |
| **Technology & data Variations** | None | |

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| **Alternative flow** | Reset: Invalid password | |
| **Release (id)** | UC5.3, v0.1, 25/04/2026 | |
| **Context of Use** | Account management. | |
| **Scope** | The system alerts the anonymous user that they have inserted an invalid password. | |
| **Level** | Subfunction | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to reset password |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide. | |
| **Minimal Guarantees** | No unsecure password is created. | |
| **Success Guarantees** | None | |
| **Trigger** | UC5 | |
| **Description (Main Success Scenario)** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid password, specifying the problem. |
| | 2 | The sequence of events starts at step 12 of the main flow (UC5). |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | Login | |
| **Release (id)** | UC6, v0.1, 20/04/2026 | |
| **Context of Use** | User authentication. | |
| **Scope** | Let an anonymous user login. | |
| **Level** | Primary task | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to login |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be logged out. | |
| **Minimal Guarantees** | No user login in an account they have no authorization for. | |
| **Success Guarantees** | The anonymous user is logged in her account. | |
| **Trigger** | The anonymous user selects “login”. | |
| **Description** | **Step** | **Action** |
| | 1 | The anonymous user selects the functionality “login”. |
| | 2 | The system asks the user to insert username and password. |
| | 3 | The user inserts the required data. |
| | 4 | The system checks whether both the username and password correspond to an existing account. |
| | 5 | The system switches to the authenticated user functionalities. |
| **Extension** | **Step** | **Other action** |
| | 4a | Invalid username or password: UC5.1 |

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| **Technology & data Variations** | None |

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|---|---|---|
| **Alternative flow** | Login: Invalid username or password | |
| **Release (id)** | UC6.1, v0.2, 25/04/2026 | |
| **Context of Use** | User authentication. | |
| **Scope** | The system alerts the anonymous user that they have inserted an invalid username or password. | |
| **Level** | Subfunction | |
| **Primary actor** | Anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user | Wants to login |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user inserted a pair of username and password that does not correspond to any existing account | |
| **Minimal Guarantees** | No user login in an account they have no authorization for. | |
| **Success Guarantees** | None | |
| **Trigger** | UC5 | |
| **Description (Main Success Scenario)** | **Step** | **Action** |
| | 1 | The system alerts the user that has inserted an invalid username or password, specifying the problem. |
| | 2 | The sequence of events starts at step 2 of the main flow (UC5). |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | Logout | |
| **Release (id)** | UC7, v0.1, 20/04/2026 | |
| **Context of Use** | User authentication. | |
| **Scope** | Let an authenticated user logout. | |
| **Level** | Primary task | |
| **Primary actor** | Authenticated user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Authenticated user | Wants to logout |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be logged in. | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The authenticated user is logged out. | |
| **Trigger** | The anonymous user selects “logout”. | |
| **Description** | **Step** | **Action** |
| | 1 | The anonymous user selects the functionality “logout”. |
| | 5 | The system switches to the anonymous user functionalities. |
| **Extension** | **Step** | **Other action** |
| | - | None |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | Ticket display | |
| **Release (id)** | UC8, v0.1, 21/04/2026 | |
| **Context of Use** | Ticketing management. | |
| **Scope** | Let a user view the details of a ticket. | |
| **Level** | Primary task | |
| **Primary actor** | Authenticated user (passenger, train company or network manager) or anonymous user | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Authenticated user | Wants to view tickets' details |
| | Anonymous user | Wants to view tickets' details |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be in an interface where a ticket is displayed.<br>Magari da ultra-specificare gli include<br>Arrivi da: path, train, station | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The user is in an interface displaying the ticket's details. | |
| **Trigger** | The user selects "view tickets' details". | |
| **Description** | **Step** | **Action** |
| | 1 | The authenticated user selects the functionality "view tickets' details". |
| | 2 | The system displays the ticket's details. |
| **Extension** | **Step** | **Other action** |
| | - | None |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | Ticket purchase | |
| **Release (id)** | UC9, v0.2, 25/04/2026 | |
| **Context of Use** | Ticketing management. | |
| **Scope** | Enable a passenger to purchase a ticket. | |
| **Level** | Primary task | |
| **Primary actor** | Passenger | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Passenger | Wants to buy a ticket |
| | Payment gateway | Manages transaction |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The passenger must be logged in.<br>The passenger must be in an interface displaying the ticket's details. | |
| **Minimal Guarantees** | The passenger who cannot pay does not purchase any ticket. | |
| **Success Guarantees** | The passenger successfully purchased a ticket. | |
| **Trigger** | The passenger selects "Buy ticket". | |
| **Description** | **Step** | **Action** |
| | 1 | The passenger selects the functionality "Buy ticket". |
| | 2 | The system sends a transaction request to the payment gateway. |
| | 3 | The payment gateway manages the transaction. |
| | 4 | The payment gateway sends the transaction's status to the system. |
| | 5 | If the transaction is not successful |

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| | 5.1 | The system alerts the passenger about the failed transaction. |
| | 6 | Otherwise |
| | 6.1 | The system alerts the passenger about the completed transaction. |
| | 6.2 | The system notifies the passenger of successful purchase with an email. |
| | 6.3 | The system saves the ticket in the account archive. |
| **Extension** | **Step** | **Other action** |
| | - | None |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | Ticket cancellation | |
| **Release (id)** | UC10, v0.1, 21/04/2026 | |
| **Context of Use** | Ticketing management. | |
| **Scope** | Enable a passenger to cancel a purchased ticket. | |
| **Level** | Primary task | |
| **Primary actor** | Passenger | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Passenger | Wants to buy a ticket |
| | Payment gateway | Manages transaction |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The passenger must be logged in.<br>The passenger must be in an interface displaying the ticket's details.<br>The passenger must have already purchased the ticket. | |
| **Minimal Guarantees** | The passenger does not cancel any ticket they have not purchased. | |
| **Success Guarantees** | The passenger successfully cancelled a purchased ticket. | |
| **Trigger** | The passenger selects "Cancel ticket". | |
| **Description** | **Step** | **Action** |
| | 1 | The passenger selects the functionality "Cancel ticket". |
| | 2 | The system sends a transaction request to the payment gateway. |
| | 3 | The payment gateway manages the transaction. |
| | 4 | The payment gateway sends the transaction's status to the system. |
| | 5 | If the transaction is not successful |

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| | 5.1 | The system alerts the passenger about the failed transaction. |
| | 6 | Otherwise |
| | 6.1 | The system alerts the passenger about the completed transaction. |
| | 6.2 | The system notifies the passenger of successful purchase with an email. |
| | 6.3 | The system saves the ticket's cancellation information in the account archive. |
| **Extension** | **Step** | **Other action** |
| | - | None |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | Ticket history | |
| **Release (id)** | UC11, v0.1, 21/04/2026 | |
| **Context of Use** | Ticketing management. | |
| **Scope** | Enable a passenger to view her tickets' history, both purchased and cancelled. | |
| **Level** | Primary task | |
| **Primary actor** | Passenger | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Passenger | Wants to view tickets' history |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The passenger must be logged in. | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The passenger successfully view tickets' history. | |
| **Trigger** | The passenger selects "Tickets' history". | |
| **Description** | **Step** | **Action** |
| | 1 | The passenger selects the functionality "Tickets' history". |
| | 2 | The system shows the list of tickets purchased and cancelled by that passenger. |
| **Extension** | **Step** | **Other action** |
| | - | None |
| **Technology & data Variations** | None | |

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| **Use case** | Find Path | |
| **Release (id)** | UC12, v0.1, 25/04/2026 | |
| **Context of Use** | Path finding | |
| **Scope** | Let an anonymous user or passenger find a path between two stations | |
| **Level** | Primary task | |
| **Primary actor** | Anonymous user OR Passenger | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Anonymous user or Passenger | Find a path |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be either logged out or logged in as a Passenger | |
| **Minimal Guarantees** | No impossible path is proposed | |
| **Success Guarantees** | The user has a suitable path and can directly select other operations for it. | |
| **Trigger** | The user selects “find a path”. | |
| **Description** | **Step** | **Action** |
| | 1 | The user selects the functionality “find a path”. |
| | 2 | The system asks the user to insert a Departing and Arriving station |
| | 3 | The user inserts all the requested data. |
| | 4 | The system ask the user an Departure or Arrival time |
| | 5 | The user provides any of the requested data, possibly confirming zero |

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| | 6 | The system checks whether at least one path along the mesh between the two stations exists |
| | 7 | The system computes whether one or the intersection of multiple service lines (referred to as candidate paths) connects the stations |
| | 8 | The system filters out all candidate paths that do not respect Departure and Arrival times, if provided |
| | 9 | The system presents the user an option to change path-sorting algorithm from a default option |
| | 10 | The system uses the path-sorting algorithm to provide the top-ranking paths computed at step 8 |
| | 11 | The user selects one path |
| | 12 | The system presents to the user all path-specific actions regarding that path (UC 12.1) |
| **Extension** | **Step** | **Other action** |
| | 3a | If the user does not insert sufficient data (leaves arrival or departure station blank), display “no arrival/departure station” error and return to step 2 |
| | 5a | If the user inserts zero data, default to “Departure” as current time, proceed normally |
| | 6a | If no path is found, produce “Stations disconnected” error and return to step 2 |
| | 7a | If no candidate paths are found, produce “No path available” error and return to step 2 |
| | 8a | If no candidate paths remain after the filter, display “Too strict Arrival/Departure” error and return to step 4 |
| | 9-10a | If the user changes sorting algorithm, the newly selected one is used to repeat from step 8 |

| | | |
|---|---|---|
| | 9-10b | If two or more paths score equally in selected sorting algorithm, use all other sorting algorithms in a pre-determined order as tie-breakers, and failing that, assign order among them randomly.<br><br>Proceed as normal |
| **Technology & data Variations** | If user is not a logged-in Passenger, ticket actions for the path are displayed but marked as unavailable | |

| | | |
|---|---|---|
| **Use case** | Purchase Tickets in path | |
| **Release (id)** | UC12.1, v0.1, 28/04/2026 | |
| **Context of Use** | Path finding, Ticket Purchase | |
| **Scope** | Let a Passenger directly purchase tickets involved in path | |
| **Level** | Subfunction | |
| **Primary actor** | Passenger | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Passenger | Purchase tickets |
| | Payment Gateway | Process transaction |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The user must be either logged in as a Passenger | |
| **Minimal Guarantees** | No unnecessary tickets are proposed | |
| **Success Guarantees** | The user purchased each ticket | |
| **Trigger** | The user selects “Tickets” at step 13 of UC12 | |
| **Description** | **Step** | **Action** |
| | 1 | The Passenger selects “Tickets” |
| | 2 | The system computes the cheapest set of tickets that cover transit in all trains of the path for the duration of the path itself |
| | 3 | All tickets already owned by the passenger are marked as “Owned” on the list. |
| | 4 | The system provides such list to the user, along with the option to include(“UC8 Ticket Display”) for each.<br><br>If all tickets are marked as “Owned”, jump to step 6 |

| | | |
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| | 5 | Upon UC8 completion (both with purchase or not) in step 4, return to step 3. |
| | 6 | The system displays a “All tickets owned!” message<br><br>Return to UC12 Step 3, marking “Tickets” option as complete |
| **Extension** | **Step** | **Other action** |
| | 2a | If no set of tickets exist, produce “Ticketing Impossible” error, return to UC12 step 13, marking “Tickets” option as error |
| | 4a | If a well-formed UC8 option for a ticket cannot be made, and it is not already marked “Owned”, provide “Invalid Ticket” error and mark that option as error.<br><br>Continue normally |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | View Info | |
| **Release (id)** | UC13, v0.1, 28/04/2026 | |
| **Context of Use** | Info Display | |
| **Scope** | Let any user view information of trains and stations | |
| **Level** | Subfunction | |
| **Primary actor** | User | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | User | View detailed information |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | None | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The user views the train service's or station's data | |
| **Trigger** | The user selects an abbreviated form of station or train service information | |
| **Description** | **Step** | **Action** |
| | 1 | The Passenger selects the abbreaviated information |
| | 2A | The system displays the station's details |
| | 3A | The system displays an abbreviated form of train information for all trains passing through the station |
| | 4A | If the user is a Passenger, the system also computes and displays options for tickets departing from the station |
| | 2B | The system displays the train's details |

| | | |
|---|---|---|
| | 3B | The system displays an abbreviated form of station information for all stations the train passes through |
| | 4B | If the user is a Passenger, it displays options for tickets for the train |
| | 5 | The user may select any abbreviated information to expand into another UC13 View Information |
| | 6 | Passengers may select displayed tickets to expand into U8 Ticket Display |
| **Extension** | **Step** | **Other action** |
| | | None |
| **Technology & data Variations** | If a the information belongs to a station, the path A is chosen. Otherwise, if the information belongs to a train service, path B is chosen. | |

| | | |
|---|---|---|
| **Use case** | Search Info | |
| **Release (id)** | UC14, v0.1, 28/04/2026 | |
| **Context of Use** | Info Display | |
| **Scope** | Let any user find station by name or Train service by ID | |
| **Level** | Primary Task | |
| **Primary actor** | User | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | User | Find information by query |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | None | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The User is given a list of matching results, or is warned that there are no matches otherwise | |
| **Trigger** | User selects “Search” | |
| | **Step** | **Action** |
| | 1 | The user selects “Search” |
| | 2 | The system asks the user Station name or train ID for the query |
| | 3 | The user provides a string |
| | 4 | The system matches that string agains all known names of stations and train service IDs |
| | 5 | For all matching stations/trains, a reduced set of their information is prepared and appended to a return list |
| | 6 | The system provides the user the return list |

| | | |
|---|---|---|
| | 7 | The user may select the abbrevited information to expand to a full UC13 Info View |
| **Extension** | **Step** | **Other action** |
| | 4a | If no name matches the string, produce “No match” error and return to step 2 |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | Show Map | |
| **Release (id)** | UC15, v0.1, 28/04/2026 | |
| **Context of Use** | Info Display | |
| **Scope** | Let any user see the stations and their connections | |
| **Level** | Primary Task | |
| **Primary actor** | User | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | User | See topology |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | None | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The User is able to see stations distributed according to the topology of the network, and select one to expand into UC13 | |
| **Trigger** | User selects “Show Map” | |
| **Description** | **Step** | **Action** |
| | 1 | The user selects “Show Map” |
| | 2 | The system displays the topological map |
| | 3 | If the user selects a station, initiate UC13 for that station |
| **Extension** | **Step** | **Other action** |
| | | None |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Insert Station Data** | |
| **Release (id)** | UC17, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the System provide a standardised way to insert a Station's data | |
| **Level** | Subfunction | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to insert Station's Data |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | None | |
| **Minimal Guarantees** | No invalid Station is returned to the calling subsystem | |
| **Success Guarantees** | A valid Station is returned to the calling subsystem | |
| **Trigger** | Another subsystem includes this use case | |
| **Description** | **Step** | **Action** |
| | 1 | The system asks the user to insert:<br>• The Station's name<br>• The Station's coordinates<br>• The Station's platforms, including:<br>  ○ their locally unique identifier<br>  ▪ e.g. "1" or "1-East" |
| | 2 | The user inserts the requested data. |
| | 3 | The user submits the data |
| | 4 | The created station is returned to the calling subsystem |
| **Extensions** | **Step** | **Alternate Action** |
| | 3a | If any of the following checks fail:<br>• The Station's name must be unique<br>  ○ Also amongst Junction Names<br>• All platforms identifiers must be unique<br>  ○ Within a single station |
| | 3a.1 | Display an error |

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| | 3a.2 | Return to step 1 |
| **Technology & data Variations** | If a base Station is provided by the calling subsystem<br>• fields should be pre-populated with the base Station’s data | |

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|---|---|---|
| **Use case** | **Insert Junction Data** | |
| **Release (id)** | UC18, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the System provide a standardised way to insert a Junction's data | |
| **Level** | Subfunction | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to insert Junction's Data |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | None | |
| **Minimal Guarantees** | No invalid Junction is returned to the calling subsystem | |
| **Success Guarantees** | A valid Junction is returned to the calling subsystem | |
| **Trigger** | Another subsystem includes this use case | |
| **Description** | **Step** | **Action** |
| | 1 | The system asks the user to insert:<br>• The Junction's name<br>• The Junction's Coordinates |
| | 2 | The user inserts the requested data. |
| | 3 | The user submits the data |
| | 4 | The Junction is returned to the calling subsystem |
| **Extensions** | **Step** | **Alternate action** |
| | 3a | If any of the following are false:<br>• The Junction's name must be unique<br>  ○ Also amongst Station Names |
| | 3a.1 | Display an error |
| | 3a.2 | Go to step 1 |
| **Technology & data Variations** | If a base Junction is provided by the calling subsystem<br>• Fields should be pre-populated with the base Junction's data. | |

| | | |
|---|---|---|
| **Use case** | **Insert Line Data** | |
| **Release (id)** | UC19, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the System provide a standardised way to insert a Line's data | |
| **Level** | Subfunction | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to insert Line's Data |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | None | |
| **Minimal Guarantees** | No invalid Line is returned to the calling subsystem | |
| **Success Guarantees** | A valid Line is returned to the calling subsystem | |
| **Trigger** | Another subsystem includes this use case | |
| **Description** | **Step** | **Action** |
| | 1 | The system asks the user to insert:<br>• The Line's origin Junction / Station<br>• The Line's destination Junction / Station<br>• The Line's number of tracks<br>• The Line's length<br>• The Line's maximum speed |
| | 2 | The user inserts the requested data. |
| | 3 | The user submits the data |
| | 4 | The system returns the Line to the calling subsystem |
| **Extensions** | **Step** | **Other Action** |
| | 3a | If any of the following conditions fail:<br>• The Line's origin must exist<br>• The Line's destination must exist |
| | 3a.1 | Display an error |
| | 3a.2 | Go to step 1 |
| **Technology & data** | If a base Line is provided by the calling subsystem, | |

| | | |
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| **Variations** | • Fields should be pre-populated. | |
| **Use case** | **Create Station** | |
| **Release (id)** | UC20, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager create a Station | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to create a Station |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | No invalid Station is registered | |
| **Success Guarantees** | A Station is registered | |
| **Trigger** | The Network Manager selects “Register a station”. | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager selects “Register a station”. |
| | 2 | Include(Insert Station Data) |
| | 3 | The system registers the Station. |
| **Extensions** | **Step** | **Other Action** |
| | - | None |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Create Junction** | |
| **Release (id)** | UC21, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager create a Junction | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to create a Junction |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | No invalid Junction is registered | |
| **Success Guarantees** | A Junction is registered | |
| **Trigger** | The Network Manager selects ”Register a junction”. | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager selects ”Register a junction”. |
| | 2 | Include(Insert Junction Data) |
| | 3 | The system registers the Junction. |
| **Extensions** | **Step** | **Other Action** |
| | - | None |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Create Line** | |
| **Release (id)** | UC22, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager create Lines | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to create a Line |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | No invalid Line is registered | |
| **Success Guarantees** | A Line is registered | |
| **Trigger** | The Network Manager selects “Register a line”. | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager selects “Register a line”. |
| | 2 | Include(Insert Line Data) |
| | 3 | The system registers the Line. |
| **Extensions** | **Step** | **Other Action** |
| | - | None |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Edit Station** | |
| **Release (id)** | UC23, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager edit a Station | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to edit a Station |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | No service information is invalidated | |
| **Success Guarantees** | The selected Station is edited | |
| **Trigger** | The Network Manager selects “Edit station”. | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager selects “Edit station”. |
| | 2 | Include(Insert Station Data) and provide the selected Station as a base. |
| | 3 | The system edits the Station with the data provided by step 2. |
| **Extensions** | **Step** | **Other Action** |
| | 3a | If the platforms were modified and a service stops at a platform that was removed |
| | 3a.1 | Display an error to the user. |
| | 3a.2 | End |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Edit Junction** | |
| **Release (id)** | UC24, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager edit a Junction | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to edit a Junction |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | None | |
| **Success Guarantees** | The selected Junction is edited | |
| **Trigger** | The Network Manager selects “Edit junction”. | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager selects “Edit junction”. |
| | 2 | Include(Insert Junction Data) and provide the selected Junction as a base |
| | 3 | The system edits the Junction with the data provided by step 1 |
| **Extensions** | **Step** | **Other Action** |
| | - | None |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Edit Line** | |
| **Release (id)** | UC25, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager edit a Line | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to edit a Line |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | No service information is invalidated | |
| **Success Guarantees** | The selected Line is edited | |
| **Trigger** | The Network Manager selects “Edit line”. | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager selects “Edit line”. |
| | 2 | Include(Insert Line Data) and provide the selected Line as a base |
| | 3 | The system edits the Line with the data provided by step 1 |
| **Extensions** | **Step** | **Other Action** |
| | 3a | If the line’s destination or origin are modified and a service uses the line |
| | 3a.1 | Display an error to the user. |
| | 3a.2 | End |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Delete Station** | |
| **Release (id)** | UC26, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager delete a station | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to delete a station |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | No service information is invalidated | |
| **Success Guarantees** | A station and it's connected lines (if any) are deleted | |
| **Trigger** | The Network Manager select "Delete station". | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager select "Delete station". |
| | 2 | The system checks if any Service is using this station |
| | 3 | The system checks If any Line connects to this station |
| | 4 | The station is deleted |
| **Extensions** | **Step** | **Other Action** |
| | 2a | If any service is scheduled to use this station |
| | 2a.1 | Then display an error informing the user that the station cannot be deleted as it is in use |
| | 2a.2 | End |
| | 3a | If any line is connected to the station |

| | | |
|---|---|---|
| | 3a.1 | Then display a warning that informs the user of the lines that will subsequently be deleted and allows them to continue or abort the deletion. |
| | 3a.2 | If the user selects abort |
| | 3a.2.1 | End |
| | 3a.3 | Otherwise |
| | 3a.3.1 | Foreach connected line |
| | 3a.3.1.1 | Cascade Include(Delete Line) |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Delete Junction** | |
| **Release (id)** | UC27, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager Delete a junction | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to delete a junction |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | No service information is invalidated | |
| **Success Guarantees** | A junction and it's connected lines (if any) are deleted | |
| **Trigger** | The Network Manager selects "delete junction". | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager selects "delete junction". |
| | 2 | The system checks if any Service is using this junction |
| | 3 | The system checks if any Line connects to this junction |
| | 4 | The junction is deleted |
| **Extensions** | **Step** | **Other Action** |
| | 2a | If any service is scheduled to use the junction |
| | 2a.1 | Then display a warning that informs the user of the impacted service(s), allowing them to choose to continue or abort the deletion. |
| | 2a.2.1 | If they chose to continue, End |
| | 2a.2.2 | Else Go to step 3 |
| | 3a | If any line is connected to the junction |
| | 3a.1 | Then display a warning that informs the user of the lines that will subsequently be deleted and allows them to choose to continue or abort the deletion. |
| | 3a.2 | If the selects abort |
| | 3a.2.1 | End |

| | | |
|---|---|---|
| | 3a.3 | Otherwise |
| | 3a.3.1 | Foreach connected line |
| | 3a.3.1.1 | Cascade Include (Delete Line) |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Delete Line** | |
| **Release (id)** | UC28, v0.3, 26/04/2026 | |
| **Context of Use** | Network Management | |
| **Scope** | Let the network manager delete a line | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Network Manager | Wants to delete a Line |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated | |
| **Minimal Guarantees** | No service information is invalidated | |
| **Success Guarantees** | A valid Line is deleted | |
| **Trigger** | The Network Manager selects “delete line”. | |
| **Description** | **Step** | **Action** |
| | 1 | The Network Manager selects “delete line”. |
| | 2 | The system checks if any Service is scheduled to use the Line |
| | 3 | The line is deleted |
| **Extensions** | **Step** | **Other Action** |
| | 2a | If any service is scheduled to use the line |
| | 2a.1 | Then display a warning that informs the user of the impacted service(s), allowing them to choose to continue or abort the deletion. |
| | 2a.2 | If they chose to continue |
| | 2a.2.1 | Go to step 3 |
| | 2a.2.2 | Else, End |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Defining a Train Type** | |
| **Release (id)** | UC29, v0.3, 30/04/2026 | |
| **Context of Use** | Service Management | |
| **Scope** | Let a Train Company register one of their trains for use in services | |
| **Level** | Primary task | |
| **Primary actor** | Train Company | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Train Company | Wants to register one of their trains |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Train Company is authenticated | |
| **Minimal Guarantees** | No duplicated train identifier is registered | |
| **Success Guarantees** | A train type is registered | |
| **Trigger** | The Train Company initiates the creation of a Train Type. | |
| **Description** | **Step** | **Action** |
| | 1 | The Train Company initiates the creation of a Train Type. |
| | 2 | The system asks the user to insert:<br>• The Train Type's identifier<br>• The Train Type's max speed<br>• The Train Type's type (Cargo / Passenger)<br>• If Passenger<br>  ○ The Train Type's seated Capacity<br>  ○ The Train Type's standing Capacity |
| | 3 | The user submits the requested data. |
| | 4 | The train is registered. |
| **Extensions** | **Step** | **Other Action** |
| | 3a | If the Train Type identifier is not unique |
| | 3a.1 | Then display an error informing the user. |
| | 3a.2 | Return to step 2 |
| **Technology & data Variations** | None | |

| | | |
|---|---|---|
| **Use case** | **Registering a Service Type** | |
| **Release (id)** | UC30, v0.4, 03/05/2026 | |
| **Context of Use** | Service Management | |
| **Scope** | Let a Train Company register a service type | |
| **Level** | Primary task | |
| **Primary actor** | Train Company | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Train Company | Wants to register a service Type |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Train Company is authenticated | |
| **Minimal Guarantees** | No duplicated commercial name is registered | |
| **Success Guarantees** | A service is registered | |
| **Trigger** | The Train Company initiates the creation of a Service Type. | |
| **Description** | **Step** | **Action** |
| | 1 | The Train Company initiates the creation of a Service Type |
| | 2 | The system shows a list of registered Train Types. |
| | 3 | The system asks the user to select an existing or create a new Train Type. |
| | 4 | The user selects the Train Type. |
| | 5 | The system asks the user to insert:<br>• This Service Type's type (Passenger / Cargo)<br>• The service's commercial name<br>  ○ e.g. "Regionale" / "Frecciarossa"<br>• The service's price / km |
| | 6 | The user submits the requested data. |
| | 7 | The service type is registered. |
| **Extensions** | **Step** | **Other Action** |
| | 4a | If the user decides to create a new Train Type |
| | 4a.1 | Then (Defining a Train Type). |
| | 4a.2 | Select the newly created Train Type. |
| | 6a | If the service's commercial name is not unique |

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| | 6a.1 | Then display an error informing the user. |
| | 6a.2 | Return to step 4 |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | **Service Request Creation** | |
| **Release (id)** | UC31, v0.1, 03/05/2026 | |
| **Context of Use** | Service Management | |
| **Scope** | Let a Train Company create a service request | |
| **Level** | Primary task | |
| **Primary actor** | Train Company | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Train Company | Wants to create a service request |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Train Company is authenticated | |
| **Minimal Guarantees** | No invalid service is created | |
| **Success Guarantees** | A service is created | |
| **Trigger** | The Train Company initiates the creation of a Service Request. | |
| **Description** | **Step** | **Action** |
| | 1 | The Train Company initiates the creation of a Service Request. |
| | 2 | The system shows a list of registered Service Types. |
| | 3 | The system asks the user to select an existing or create a new Service Type. |
| | 4 | The user selects the Service Type. |
| | 5 | The system asks the user to insert:<br>• A list of destinations as:<br>  ○ Stations for the service to stop at / go via<br>  ○ Junctions for the service to go via<br>• The time taken between destinations<br>• A list of Dispatch Times (Applies to the first station in the list):<br>  ○ A time in terms of `hh:mm`<br>  ○ Train number<br>  ○ On what days of the week the dispatch time is used |

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| | 6 | The user submits the requested data. |
| | 7 | The service request is created. |
| **Extensions** | **Step** | **Other Action** |
| | 4a | If the user decides to create a new Service Type |
| | 4a.1 | Then (Defining a Train Type) |
| | 4a.2 | Select the newly created Train Type. |
| | 6a.1 | If any of the following are true:<br>• The user did not insert at least 2 stations in the destinations list<br>• The first element of the destinations list is not a station<br>• Any train number inserted is not unique across all services |
| | 6a.1.1 | Display an error |
| | 6a.1.2 | Go to step 4 |
| **Technology & data Variations** | None | |

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|---|---|---|
| **Use case** | **Service Request Review** | |
| **Release (id)** | UC32, v0.1, 03/05/2026 | |
| **Context of Use** | Service Management | |
| **Scope** | Let the Network Manager review a service request | |
| **Level** | Primary task | |
| **Primary actor** | Network Manager | |
| **Stakeholder & interest** | **Stakeholder** | **Interest** |
| | Train Company | Wants to review a service request |
| | System | Manages the user interface, the database and their interactions |
| **Preconditions** | The Network Manager is authenticated.<br>There is a Service Request not yet reviewed. | |
| **Minimal Guarantees** | An invalid service request is denied. | |
| **Success Guarantees** | A service is approved, with possible amendment. | |
| **Trigger** | The Network Management initiates the review of a Service Request. | |
| **Description** | **Step** | **Action** |
| | 1 | The Train Company initiates the review of a Service Request. |
| | 2 | The system shows a list of registered Service Requests. |
| | 3 | The system asks the user to select a Service Request. |
| | 4 | The user selects a Service Request. |
| | 5 | The system performs automated service integrity checks. |
| | 6 | The system shows the results of the integrity checks. |
| | 7 | The system asks the user to select an operation between deny, amend and approve. |
| | 8 | If the user selects “deny” |
| | 8.1 | The system deletes the service request. |
| | 9 | If the user selects “amend” |
| | 9.2 | The system shows data specified at UC31[step5]. |
| | 9.3 | The network manager inserts changes. |
| | 9.4 | Go to step 5 |
| | 10.1 | If the user selects “approve” |
| | 10.2 | The system saves the service as effective. |

| | | |
|---|---|---|
| | 11 | The system notifies the Train Company that requested the service. |
| **Extensions** | **Step** | **Other Action** |
| | 9.3a.1 | If any of the following are true:<br>• The user did not insert at least 2 stations in the destinations list<br>• The first element of the destinations list is not a station<br>Any train number inserted is not unique across all services |
| | 9.3a.1.1 | Display an error |
| | 9.3a.1.2 | Go to step 9.2 |
| **Technology & data Variations** | None | |